

Neurodiversity-Informed AI Agent for an Inclusive Medical VR Training Environment

Project Overview

Virtual Reality (VR) is increasingly used in healthcare education to provide students with realistic and immersive environments in which they can practise clinical skills. However, VR environments can also create significant cognitive and sensory demands through visual complexity, background noise, time pressure, and intensive interaction. These factors may particularly affect neurodivergent learners or users with different sensory and accessibility needs.

The objective of this project is to design and develop a functional medical VR training prototype incorporating an intelligent virtual agent that can assist users and adapt aspects of the virtual environment according to their individual needs. The system should demonstrate how AI and adaptive VR technologies can be combined to create a more inclusive, accessible, and psychologically safe learning environment.

The project will be undertaken by a team of approximately five students and should result in a working prototype suitable for demonstration and future user evaluation.

System Requirements

The system should provide a multiplayer-capable VR medical training environment containing a simulated clinical scenario. A multi-casualty trauma scenario is proposed, although the scope of the medical scenario can be simplified depending on development time and available resources.

A central component of the system will be an AI-powered virtual agent embedded within the VR environment. Users should be able to interact with the agent during the simulation using natural language, through voice and/or text. The agent should be capable of:

- guiding users through the medical training scenario and providing contextual instructions;
- answering questions related to the current task or simulation;
- communicating information in clear and supportive ways;
- adapting the style, amount, or complexity of information presented to the user; and
- recommending environmental adjustments when appropriate.

The AI functionality may be implemented using an appropriate Large Language Model (LLM) service, such as Azure OpenAI or another suitable conversational AI platform.

Adaptive and Accessibility Features

The system should allow users to customise the VR experience according to their sensory and interaction preferences. The prototype should implement several adaptive features, such as:

- reducing or controlling background sounds;
- adjusting environmental lighting;
- modifying colours or visual contrast;
- reducing unnecessary visual stimuli;
- providing a simplified interface or interaction mode;
- controlling the amount or pace of information presented by the AI agent; and
- providing virtual sensory-regulation objects or tools with which users can interact.

These adaptations may be activated manually through a user-accessible settings interface and/or recommended by the AI agent during the simulation.

AI Agent and User Interaction

The virtual agent should be represented as a character within the VR environment rather than only as a conventional chatbot interface. The agent should demonstrate inclusive and neurodiversity-informed communication without relying on stereotypical representations.

Where technically feasible, the system should maintain contextual awareness of the simulation—for example, knowing the user's current stage or task—and use this information when responding to questions or providing assistance.

The architecture should be designed so that different agent behaviours or configurations can be supported in the future. For example, future researchers should be able to compare a standard educational AI agent with a neurodiversity-informed adaptive agent without rebuilding the complete VR application.

Expected Deliverables

By the end of the project, the student team should deliver a functional VR prototype demonstrating the medical simulation, interactive AI virtual agent, and adaptive sensory/accessibility features. The project should also include the source code, system and software architecture documentation, installation/deployment instructions, and a short user guide.

The prototype does not need to implement a complete medical training platform. The priority is to build a robust and extensible proof-of-concept that successfully demonstrates the integration of VR medical simulation + conversational AI agent + adaptive sensory and accessibility features and can subsequently be extended and evaluated as part of a larger research project.